



ENGINEERING MATHEMATICS-II

Properties of Fourier Transform -2

Latha N,
Assistant Professor of Mathematics,
Dept. of Science and Humanities.

Fourier Transform of Derivatives

Let $f(t)$ be continuous and $f^{(k)}(t)$, $k = 1, 2, \dots, n$ be piecewise continuous on every interval $[-l, l]$ and

$\int_{-\infty}^{\infty} |f^{(k-1)}(t)| dt$, $k = 1, 2, \dots, n$ converges. Let $f^{(k)}(t) \rightarrow 0$ as $t \rightarrow \infty$ for $k = 0, 1, 2, \dots, n - 1$. If $\mathcal{F}[f(t)] = F(\omega)$, then

$$\mathcal{F}[f^{(n)}(t)] = (i\omega)^n F(\omega),$$

Where $f(t)$ and all its derivatives vanish at infinity.

For Example,

$$\mathcal{F}[f'(t)] = \int_{-\infty}^{\infty} f'(t)e^{-i\omega t} dt$$

$$= \left[f(t)e^{-i\omega t} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} i\omega f(t)e^{-i\omega t} dt = i\omega F(\omega).$$

$$\mathcal{F}[f''(t)] = -\omega^2 F(\omega).$$

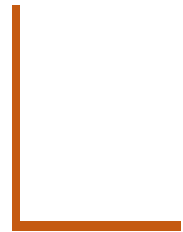
Example

Solve differential equation using Fourier transform

$$f' - 4f = u(t)e^{-4t}$$

$$i\omega F(\omega) - 4F(\omega) = \frac{1}{4 + i\omega}$$

$$(i\omega - 4)F(\omega) = \frac{1}{4 + i\omega}$$



$$F(\omega) = \frac{1}{(4 + i\omega)(i\omega - 4)}$$

$$F(\omega) = \frac{-1}{(4 + i\omega)(4 - i\omega)}$$

$$F(\omega) = -\frac{1}{4^2 + \omega^2} \quad \left\{ \frac{2a}{a^2 + \omega^2} = \mathcal{F}[e^{-a|t|}] \right\}$$

$$f(t) = -\frac{1}{2 \cdot 4} e^{-4|t|} = -\frac{1}{8} e^{-4|t|}$$

Symmetry property of Fourier transforms

Let $\mathcal{F}[f(t)] = F(\omega)$. Then

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$f(t) = \mathcal{F}^{-1}[\omega] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

$$2\pi f(t) = \int_{-\infty}^{\infty} F(s) e^{ist} ds$$

$$2\pi f(-\omega) = \int_{-\infty}^{\infty} F(s) e^{-is\omega} ds \quad \text{change } s \rightarrow t$$

$$= \int_{-\infty}^{\infty} F(t) e^{-i\omega t} dt = \mathcal{F}[F(t)]$$

Example

Find Fourier transform of $\frac{1}{2\pi} \text{sinc} t$

Solution

We know that $F(\omega) = \frac{1}{\sqrt{2\pi}} \text{sinc} \omega$ for

$$f(t) = \begin{cases} 0 & t < -1 \\ 1 & -1 < t < 1 \\ 0 & t > 1 \end{cases} \quad \text{typical box wave}$$

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$\mathcal{F}\left(\frac{1}{\sqrt{2\pi}} \operatorname{sinc} t\right) = \mathcal{F}(F(t)) =$$

$$2\pi f(-\omega) = \begin{cases} 0 & -\omega < -1 \\ 2\pi & -1 < -\omega < 1 \\ 0 & -\omega > 1 \end{cases}$$

Differential with respect to frequency (ω)

If $\mathcal{F}(f(t)) = F(\omega)$ then

$$\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$$

Differentiating with respect to ω

$$F'(\omega) = \frac{d}{d\omega} \left(\int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt \right)$$


$$F'(\omega) = \int_{-\infty}^{\infty} \frac{d}{d\omega} (f(t) \cdot e^{-i\omega t}) dt$$

$$F'(\omega) = \int_{-\infty}^{\infty} -it \cdot f(t) \cdot e^{-i\omega t} dt$$
$$F'(\omega) = -i[\mathcal{F}(tf(t))]$$

Similarly

$$F''(\omega) = (-i)^2[\mathcal{F}(t^2 f(t))]$$



Therefore $\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$

$$(-i)^n t^n f(t) = \mathcal{F}^{-1}(F^{(n)}(\omega))$$

Example

Obtain Fourier transform for $[te^{-4t}u(t)]$.

solution : w.K.t $\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$

And from previous property $\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$

$$\mathcal{F}[e^{-4t}u(t)] = \frac{1}{4 + i\omega}$$

$$\mathcal{F}[te^{-4t}u(t)] = i \cdot \frac{d}{d\omega} \left[\frac{1}{4 + i\omega} \right] = i \left[-\frac{i}{(4 + i\omega)^2} \right] = \frac{1}{(4 + i\omega)^2}$$

Fourier transform of basic functions:

Unit impulse function, $f(t) = \delta(t)$

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-i\omega t} dt = \left[e^{-i\omega t} \right]_{t=0} = 1$$

$\therefore$ Impulse in time $\Rightarrow$ constant in frequency

Fourier transform of Constant function,
 $f(t) = k$

By Duality property, we have

$$\text{If } \mathcal{F}[f(t)] = F(\omega) \text{ then } \mathcal{F}[F(t)] = 2\pi f(-\omega)$$

We know that $\mathcal{F}[\delta(t)] = 1$

$$\Rightarrow \mathcal{F}[1] = 2\pi\delta(-\omega) = 2\pi\delta(\omega) \quad (\because \text{impulse function is even})$$

$$\therefore \mathcal{F}[k] = 2\pi k\delta(\omega)$$

Hence constant in time $\Rightarrow$ Impulse in frequency

1. $f(t) = e^{iat}$

By frequency shifting property, we have

$$\mathcal{F}[e^{iat}f(t)] = F(\omega - a)$$

$$\text{Also } \mathcal{F}[1] = 2\pi\delta(\omega)$$

$$\therefore \mathcal{F}[1 \cdot e^{iat}] = \mathcal{F}[e^{iat}] = 2\pi\delta(\omega - a)$$

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1. Trigonometric functions, $f(t) = \cos at$ and $f(t) = \sin at$

$$\mathcal{F}[\cos at] = \mathcal{F}\left[\frac{e^{iat} + e^{-iat}}{2}\right] = \frac{1}{2}\{\mathcal{F}[e^{iat}] + \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\cos at] = \frac{1}{2}\{2\pi\delta(\omega - a) + 2\pi\delta(\omega + a)\}$$

$$\begin{aligned}\therefore \mathcal{F}[\cos at] &= \pi\delta(\omega - a) + \pi\delta(\omega + a) \\ &= \pi\{\delta(\omega - a) + \delta(\omega + a)\}\end{aligned}$$

$$\mathcal{F}[\sin at] = \mathcal{F}\left[\frac{e^{iat} - e^{-iat}}{2i}\right] = \frac{1}{2i}\{\mathcal{F}[e^{iat}] - \mathcal{F}[e^{-iat}]\}$$

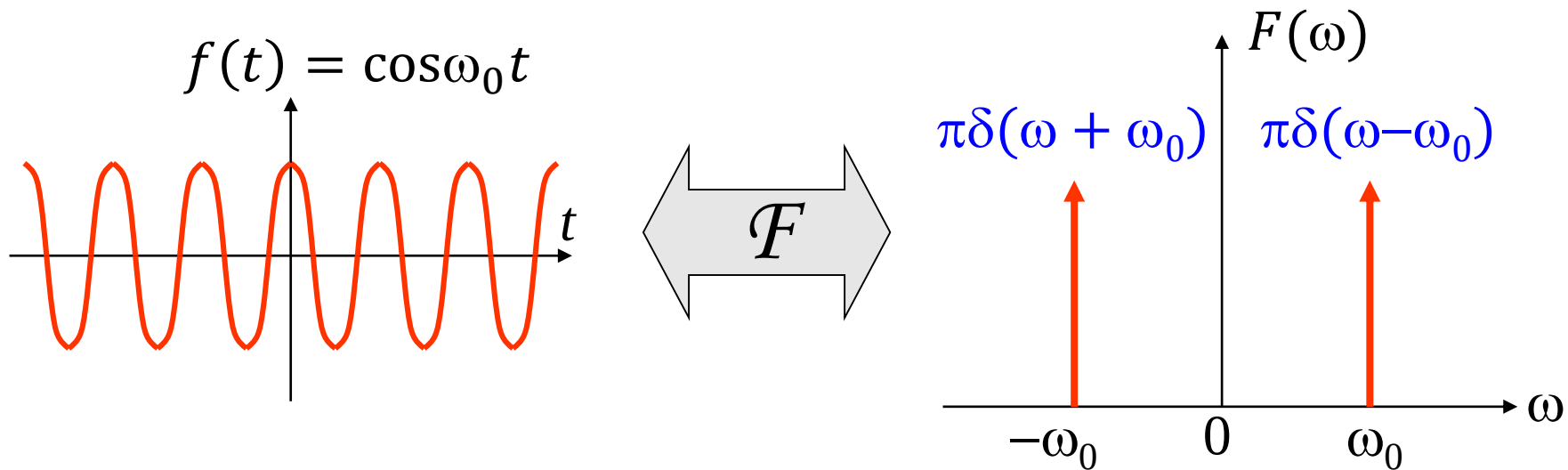
$$\begin{aligned}\mathcal{F}[\sin at] &= \frac{1}{2i}\{2\pi\delta(\omega - a) - 2\pi\delta(\omega + a)\} \\ &= \frac{1}{i}\{\pi\delta(\omega - a) - \pi\delta(\omega + a)\}\end{aligned}$$

$$\therefore \mathcal{F}[\sin at] = -i\pi\delta(\omega - a) + i\pi\delta(\omega + a) = i\pi\{\delta(\omega + a) - \delta(\omega - a)\}$$

Fourier Transforms of Sinusoidal Functions

$$\cos \omega_0 t \xleftrightarrow{\mathcal{F}} \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$$

$$\sin(\omega_0 t) \xleftrightarrow{\mathcal{F}} -i\pi \delta(\omega - \omega_0) + i\pi \delta(\omega + \omega_0)$$



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- Linearity property:

$$\mathcal{F}[a f(t) + b g(t)] = a \mathcal{F}[f(t)] + b \mathcal{F}[g(t)].$$

- Shifting property: $\mathcal{F}[f(t - a)] = e^{-i\omega a} F(\omega)$
- Frequency shifting property: $\mathcal{F}[e^{iat} f(t)] = F(\omega - a)$
- Modulation Theorem: $\mathcal{F}[f(t) \cos(\omega_0 t)] = \frac{1}{2} [F(\omega + \omega_0) + F(\omega - \omega_0)]$
- Fourier Transform of Derivatives: $\mathcal{F}[f^{(n)}(t)] = (i\omega)^n F(\omega),$
- Differential with respect to frequency (ω): $\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$
- Symmetry property $\mathcal{F}[F(t)] = 2\pi f(-\omega)$

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- $\mathcal{F}[\delta(t)] = 1$
 - $\mathcal{F}[e^{iat}] = 2\pi\delta(\omega - a)$
 - $\mathcal{F}[\cos at] = \pi\{\delta(\omega - a) + \delta(\omega + a)\}$
 - $\mathcal{F}[\sin at] = -i\pi\{\delta(\omega + a) - \delta(\omega - a)\}$
 - $\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$
 - $\mathcal{F}[e^{-a|t|}] = \frac{2a}{a^2+\omega^2}$



Thank you

Latha N,
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